MIL-C-915/28B
23 August 1976
SUPERSEDING
MIL-C-915/28A
1 August 1972

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 1000 VOLTS, TYPE SSGU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of Specification MIL-C-915.

REQUIREMENTS:
Qualification required.

Construction (watertight)

First - Copper conductor, uncoated (See Details for size).
Second - Silicone rubber - glass tape insulation (See Details for diameter).
Third - Binder (at manufacturer's option).
Fourth - Special thermoplastic jacket (gray). (See Details for thickness).
Cable surface marking required.

Note 1: When this cable is manufactured with armor, the following additional requirements shall apply:

(B) (a) The type designation shall change to SSGA.
(b) Standard thermoplastic jacket (black or gray) may be used.
(c) Braided metal armor and paint shall be added.
(d) Cable surface marking will not be required.
(B) (e) The maximum overall dimension shall be increased by 0.050-inch.
(f) Group C inspection shall include: Armor - Conformance to material, construction and coverage. When standard thermoplastic jacket is used, the air oven test does not apply.

Details

Type and size	Conductor size, Navy Standard	Diameter over insulation nominal (nom) (inch)	Jacket thickness nom (inch)	Overall diameter maximum (max) (inches)	Conductor resistance per 1000 ft at 25°Celsius (C), max (ohms)	Insulation resistance per 1000 ft minimum (min) (megohms)	Cold bending mandrel diameter max (inches)	Accelerated service loading (amps)
SSGU-50	50(19)	0.334	0.040	0.520	0.225	160	7	240
SSGU-75	75(37)	.407	.040	.602	.146	135	8	315
SSGU-100	100(61)	.453	.050	.669	.112	125	9	375
SSGU-200	200(61)	.634	.050	.872	.0560	90	11	580
SSGU-300	300(91)	.748	.050	1.001	.0375	80	13	750
SSGU-400	400(127)	.862	.050	1.118	.0268	70	14	925

(B) denotes changes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, Center Building, SEC 6124, Prince George's Center, Hyattsville, Maryland 20782 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Details (Continued)

Type and size	Conductor size, Navy Standard	Diameter over insulation nominal (nom) (inch)	Jacket thickness nom (inch)	Overall diameter maximum (max) (inches)	Conductor resistance per 1000 ft at 25°Celsius (C), max (ohms)	Insulation resistance per 1000 ft minimum (min) (megohms)	Cold bending mandrel diameter max (inches)	Accelerated service loading (amps)
SSGU-650	650(127)	1.056	0.060	1.371	0.0169	60	17	1300
SSGU-800	800(127)	1.170	.060	1.485	.0134	50	18	1550
SSGU-1000	1000(127)	1.300	.060	1.620	.0106	40	20	----
SSGU-1600	1600(127)	1.625	.075	2.010	.00668	40	25	----
SSGU-2000	2000(127)	1.810	.075	2.210	.00530	35	27	----

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Voltage withstand - Volts, root mean square (Vrms), min	5000
Insulation resistance - Megohms/1000 ft, min	(See Details)
Conductor resistance - Ohms/1000 ft, max	(See Details)
Group A:	
Visual and dimensional	
Watertightness - See MIL-C-915, table of limits for water leakage.	
Group B:	
Gas flame - 1 hour	No failure
Drip - at 95°C ± 1°C	Zero
Cold bending, cable - (See Details for mandrel diameter)	No damage
Physicals (unaged)	
Jacket (Standard thermoplastic)	
Tensile strength - pound-force per square inch (lbf/in^2) min	1800
Elongation - Percent, min	225
Jacket (Special thermoplastic)	
(B) Tensile strength - lbf/in^2, min	2300
(B) Elongation - Percent, min	250
Group C:	
Physicals (aged)	
Jacket (Standard thermoplastic)	
Hot oil immersion	
Tensile strength - lbf/in^2, min	1400
Elongation - Percent, min	50
Jacket (Special thermoplastic)	
(B) Air oven (168 hours @ 136°C)	
Tensile strength - Percent of unaged, min	80
(B) Elongation - Percent of unaged, min	60
Hot oil immersion	
(B) Tensile strength - Percent of unaged, min	75
(B) Elongation - Percent of unaged, min	50
Permanence of printing (jacket) - Cycles, min	250

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of Groups A, B and C plus the following:

Cold working (minus 20°C ± 2°C)	No damage
Gas flame (3 hours)	No failure
(B) Accelerated service - Sizes 50 through 800 (See Details for load current)	No failure

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UNIT ORDERING LENGTHS:

Requirements

Type and size	Feet (nom)
SSGU-50 through 100	1500
SSGU-200 through 1000	1000
SSGU-1600 through 2000	500

Custodians:
Army - MI
Navy - SH
Air Force - 99

Review activities:
Army - EL, AV
Navy - EC

User activities:
Army - ME, MU, WC
Navy - CG

Preparing activity:
Navy - SH

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